# CS & DA

## Probability and Statistics

## Permutations and Combinations

Lecture No. 05







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# Recap of previous lecture



**Topic**

1. Counting Principle → Addition P, Multiplication P
2. Permutation (selection & Arrangement)
3. Combination (only selection)
4. Concept of At least & At Most
5. De-Arrangements
6. Perm of Alike items.
7. Solving Methods of Various Quest.

# Topics to be Covered



**Topic**

BASICS of P & C (Part 5)

1. Circular Perm (F & 2)
2. F & 3 (Based on Perm of Alice items)
3. Summation of Numbers.

RECAP

(✗) Number of linear Arrangements of $n$ different things $= n!$

(✗) Number linear Arrangements of $n$ things in which $p$ are alike, $q$ are alike, $r$ things are alike & Rest are different =

$$= \frac{{}^{n}P_{n}}{p!q!r!} = \frac{n!}{p!q!r!}$$

where $p+q+r+\underset{(?)}{\text{Rest}} = \text{Total}(n)$

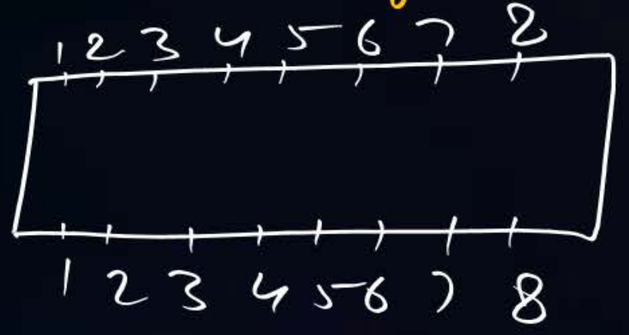
(✗) No. of De-Arrangements of $n$ things $= n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + (-1)^{n}\frac{1}{n!}\right]$

(✗) No. of Circular Permutations of $n$ different things $= \frac{n!}{n} = (n-1)!$

## Some Important Points

1. Always together (Not separated) → Assume them as one unit with in Bracket
2. All Never together (All Do not Come together) → Total − Always together
3. No two Girls are together → first arrange Boys
4. Alternately (Linear Case) → Two Cases will arise
5. Alternately (Circular Case) → only one Case will arise
6. Particular / (fix) → No Need to Select & No Need to arrange
7. Alike → use the Result $\frac{n!}{p!q!r!}$

HW8 16 persons are to be seated along a long Table with 8 chairs on each side. four particular Persons wants to be seated on one particular side while 2 persons on the other side. Then find the possible Number of Arrangements ? Ans: $(9408 \times 10!)$




Total arrangements (No Restriction) = $16!$ (Max Ans)

Required No. of arrangements (favourable arrangements)

$= ({}^8C_4 \times 4!) \times ({}^8C_2 \times 2!) \times 10! = {}^8P_4 \times {}^8P_2 \times 10!$

$$\text{Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^8P_4 \times {}^8P_2 \times 10!}{16!}$$

Analysis



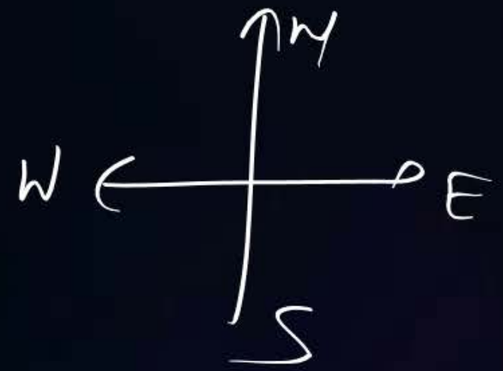

$\text{Req Ans} = {}^8P_4 \times {}^8P_2 \times 10!$

(ii) If side is not particular then Ans $= ? = ({}^2C_1 \times {}^8P_4) \times ({}^1C_1 \times {}^8P_2) \times 10!$

(M-II) Case I Facing North: $= {}^8P_4 \times {}^8P_2 \times 10!$

or

Case II (Facing South): $= {}^8P_4 \times {}^8P_2 \times 10!$



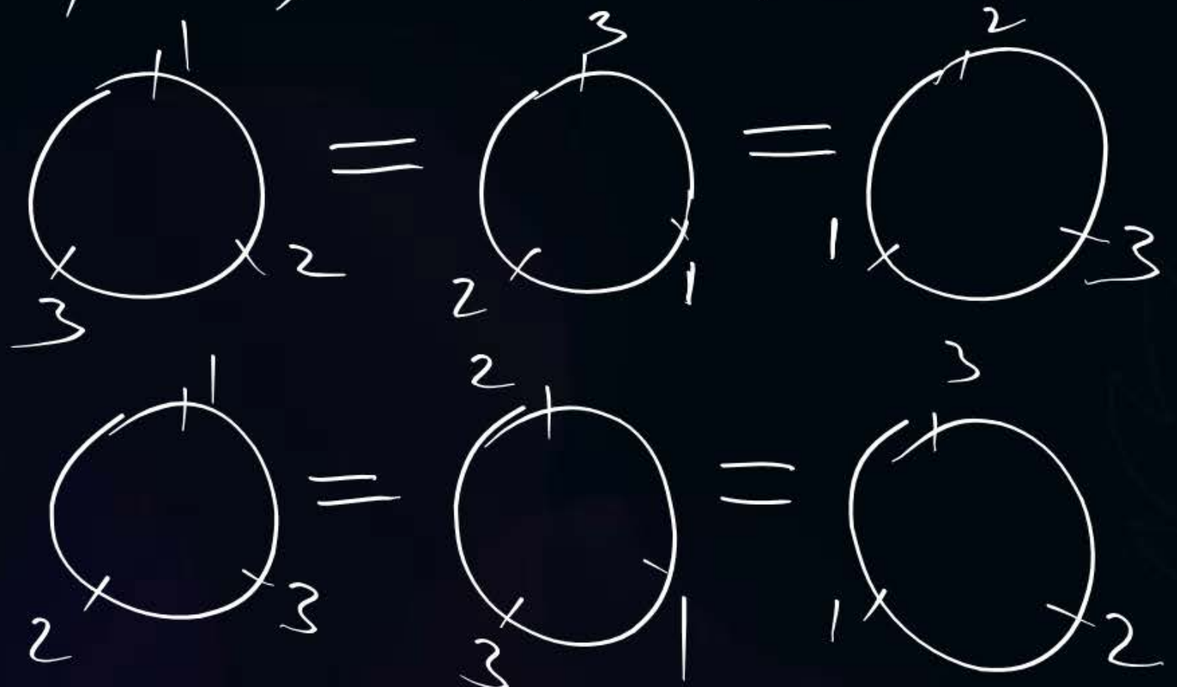

$\text{Req Ans} = 2 \times {}^8P_4 \times {}^8P_2 \times 10!$

## Circular Permutations :→

eg: For $n=3$ persons
- No. of linear arrangements $= 3! = 6$
- No of Circular arrangements $= \frac{3!}{3} = (3-1)! = 2! = 2$

CA = (123), (132), (213), (231), (312), (321) i.e 6 in arrangements

CA:

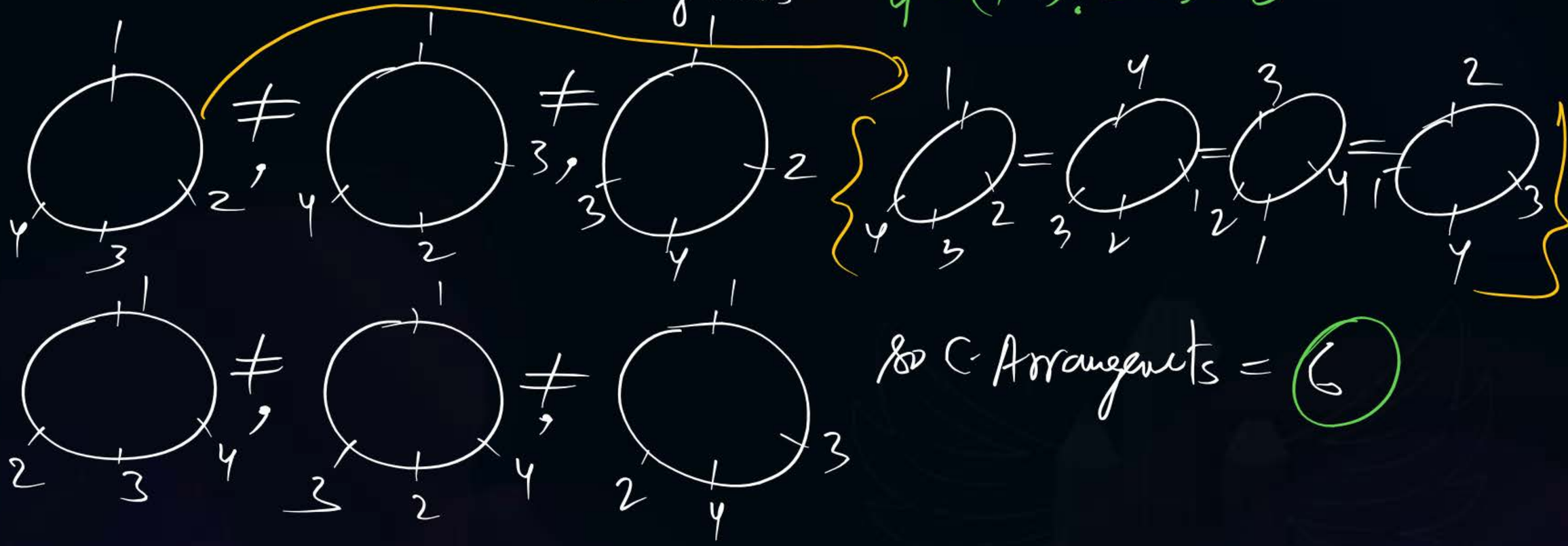


i.e Two arrangements

e.g. for $n = 4$ persons

Linear arrangements $= 4! = 24$

Circular arrangements $= \frac{4!}{4} = (4-1)! = 3! = 6$







So C. Arrangements = 6

FQ 2:- 4 Boys and 4 Girls (in which there are 2 sisters) are to be arranged around a Circular Table for Tea party then Find the possible no. of arrangements if

① there is No Restriction = ? $= (8-1)! = 7!$ (Man Ans)
(RNA)

② All G are not Separated = ? $= (5-1)! \times 4!$
Persons (Circular) Girls (Linear)




WRONG APP:- Req Ans $= (5-1)! \times (4-1)!$
(Persons) (Girls)




i.e Form Circle only once in a given Question

③ All G do not come together = ? = Total − G always together

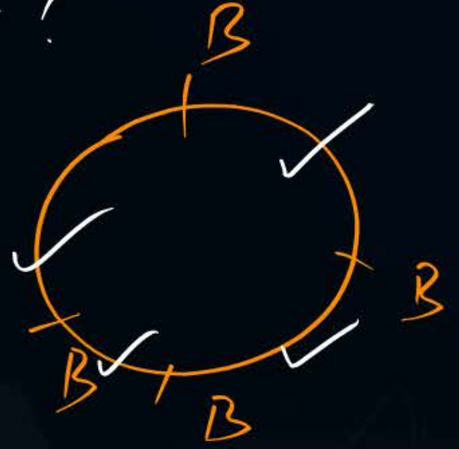
→ Never

$= (8-1)! - ((5-1)! \times 4!) = ?$

④ No two G are together = ? = First arrange Boys in a Circle

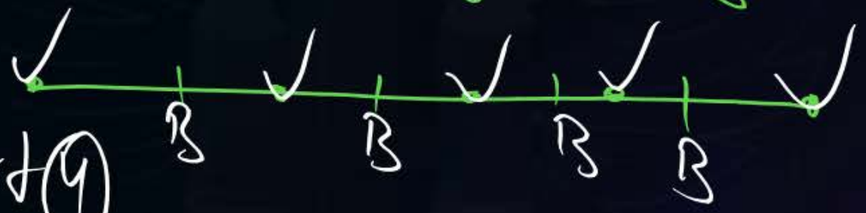
$= (4-1)! \times 4!$

(Boys) (Girls)



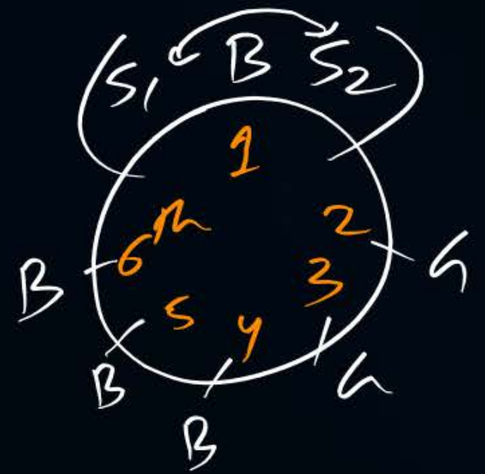

Note: while in case of linear arrangements we will get 5 fav places for Girls

⑤ B & G are to be seated alternately = ? = Same as part ④




⑥ 5G & 4B & B & G are to be seated alternately = ? = Senseless Question

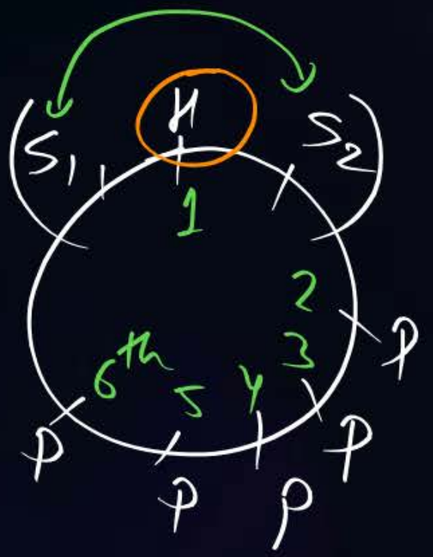
(6) there is exactly one Boy b/n two sisters = ?




$$= \left({}^{4}C_{1} \times 1\right) \times (6-1)! \times 2!$$

(Persons) (Sisters)

(7) Two sisters are to be seated at the Adjacent sides of Host ?




$$= (6-1)! \times 2!$$

No Need to Select Host bcoz Host is a particular Person & He/She knows it.

FQ3. How many 11 letter words can be formed using the letters of the word MATHEMATICS if

(1) there is No Restriction = ? $= \frac{11!}{2!2!2!}$ (= Main Ans)

M, M, A, A, T, T, E, H, I, C, S

n = 11

(2) Vowels & Consonants are not separated = ? $= 2! \times \frac{4!}{2!} \times \frac{7!}{2!2!}$

(AAEI), (MMTTHCS)

1 2

(Letters) (Vowels) (Consonants)

(3) Vowels are always together = ? $= \left(\frac{8!}{2!2!}\right) \times \frac{4!}{2!}$

M H (AAEI) C T T S M

1 2 3 4 5 6 7 8

(Letters) (Vowels)

(4) All vowels do not come together = ? = Total − Vowels always together

→ Never

$$= \left(\frac{11!}{2!\,2!\,2!}\right) - \left(\frac{8!}{2!\,2!} \times \frac{4!}{2!}\right)$$

(5) No two vowels are together = ? = first arrange consonants linearly



$$= \frac{{}^{7}P_{7}}{2!\,2!} \times \frac{{}^{8}P_{4}}{2!}$$

(Consonants) (Vowels)

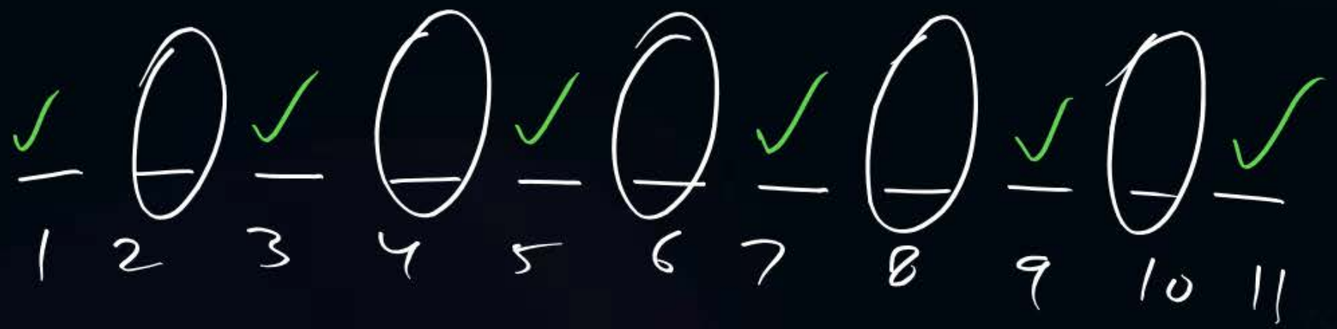
Now 8 places for vowels = 8

But vowels are only 4 (A, A, E, I)

(6) Vowels & consonants are alternately = ? = Senseless Questions

(7) Vowels occupy Even places = ?

A A E I M M I I N C S




Even place = 5 & Vowels = 4 $\Rightarrow$ Arrangement = $\frac{{}^5C_4 \times 4!}{2!} = \frac{{}^5P_4}{2!}$

Odd place = 6 & Consonants = 7

Few Places for Consonants = 6 odd places & 1 Even place

= 7 places

$= \frac{{}^5P_4}{2!} \times \frac{{}^7P_7}{2!\,2!}$

(Vowels) (Consonants)

(P8) In how many ways 7 (+) Sign & 5 (−) Sign are to be arranged in a Row so that No two (−) Sign are together ?

Sol: No two (−) Sign are together = first arrange (+) Sign

✓ + ✓ + ✓ + ✓ + ✓ + ✓ + ✓ + ✓

fav Places for (−) Sign = 8

But (−) Sign are only 5

$$= \frac{{}^{7}P_{7}}{7!} \times \frac{{}^{8}P_{5}}{5!} = 1 \times {}^{8}C_{5}$$

(+ve) (−ve)

$$= {}^{8}C_{5} = {}^{8}C_{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Note: Also find the prob in above Quest$^{n}$ ? $= \frac{\text{fav}}{\text{Total}} = \frac{56}{\left(\frac{12!}{7!5!}\right)} =$ Ans

Total arrangements (w/o Restriction) $= \frac{12!}{7! \times 5!}$

## (x) Summation of Numbers →

Sum of n digit Nos (that can be formed W/o Repetition)
(RNA)

$$= \underbrace{(\text{Sum of digits}) \times (1111\ldots\ldots1)}_{n \text{ times}} \times (n-1)!$$

eg find the sum of all 3 digit Nos that can be formed using the digits 1, 2, 3 (W/o Repetition)

Total 3 digit Nos $= 3! = 6$
(RNA)

Various Nos are

$$\begin{array}{r} 123 \\ 132 \\ 213 \\ 231 \\ 312 \\ +\ 321 \\ \hline \text{Req Sum} = 1332 \end{array}$$

(M-II) Req Sum $= (1+2+3) \times (111) \times (3-1)!$

$= 6 \times 111 \times 2 = 1332$



(2014) 2M Find the sum of 5 digit nos that can be formed using odd digits w/o Repitition.

Total 5 digit nos (RHA) $= 5! = 120$ Nos.

1, 3, 5, 7, 9

$$\text{Req Sum} = (\text{Sum of digits}) \times (11111) \times (5-1)!$$

$$= (1+3+5+7+9) \times (11111) \times 4!$$

$$= 25 \times 11111 \times 24 = 6666600$$

Q: Find the sum of all 5 digit Nos that can be formed using 0, 1, 2, 3, 4

Sol: Total 5 digit Nos = $\underset{P_1}{4\text{ways}} \times \underset{P_2}{4\text{ways}} \times \underset{P_3}{3\text{ways}} \times \underset{P_4}{2\text{ways}} \times \underset{P_5}{1\text{way}} = 96$ Nos

(RNA)

Ans: (2599980)

Case I: (if 0 is assumed as Complete Number) → then Total Nos = $5! = 120$ Nos

Req. sum of 120 Nos $= (0+1+2+3+4) \times (11111) \times (5-1)!$

$= 10 \times 11111 \times 24 = 2666640$

Case II: (Excluding 0) → Total 4 digit Nos $= 4! = 24$ Nos

Req sum of 24 Nos $= (1+2+3+4) \times (1111) \times (4-1)!$

$= 10 \times 1111 \times 6 = 66660$

Hence Sum of 96 Nos = Sum of 120 Nos − Sum of 24 Nos $= 2666640 - 66660 = 2599980$

HW8 How many 4 digit nos ($\geq 6000$) can be formed using 5, 5, 6, 6, 6, 7, 7, 7, 7

Ans = 51

(Plastic toys = RNA)

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# THANK - YOU